

Well, it all boils down to that little bit of paper and traceability for the audits. Although test certificates are based on the accuracy at the time of test, if you feel the need to have more frequent checks then the more affordable way is to have a trained calibration technician and an in-house lab.

Every R&D firm must have reference equipment against which samples are checked to have confidence in the measurement. Desai says, "It has to be a standalone reference equipment that can be used for verification. Firms producing rotating machines, tachometers and motors must have a master precision tachometer that does have valid calibration all the time. Firms producing temperature sensors must have a calibrator or master precision resistor thermometer to check against that. Mechanical gauges require master slip gauges to check

the wear and tear on instruments that are used to measure the finished goods very often."

"Test and measurement equipment distributors follow a mixed regime for calibration. They have their own calibrating equipment which are traceable to ERTL or NABL. Also, there is dependence on the respective OEM for calibration of their own brand equipment," shares Kwatra.

Importance can't be undermined

In today's world of competition, the only differentiator for your product is quality, which cannot just be assumed. To produce quality equipment, you need measurements that are accurate, reliable and repeatable.

"Not only for all the manufacturing processes but also for R&D firms, it is necessary to use calibrated equipment

to ensure reliability of results. Correctly performed calibration increases the productivity, optimises resources, and assures consistency, comparability and compatibility of products, services and acceptability," says Kela.

The importance of calibration cannot be undermined by any user. When performing any type of measurement, you must have confidence in your results that they are accurate and within specifications. Testing and measurement with calibrated equipment brings this confidence and is a form of quality assurance.

"There are many labs in India which provide calibration certificate and help maintain the accuracy of equipment. It's just the matter of awareness at all the levels," says Goliya. ●

The author is a senior technology journalist at EYF

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1
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In India (Sales, Service & Calibration)
TTL Technologies Pvt. Ltd.
#3071, 10th Cross, 11th Main,
HAL II Stage, Bangalore-560 008
Ph: (080) 25266646, 25251859
For Sales: 1800-425-3200 (Toll Free)
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